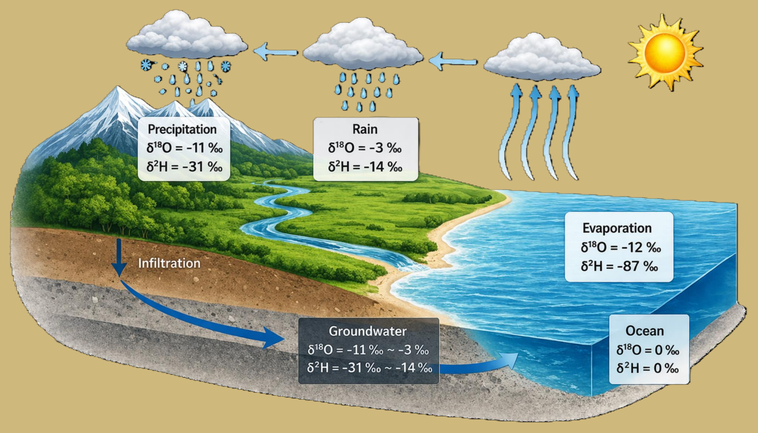
Flying Laboratory for Isotope Research (FLAIR) Graduate Projects | University of Colorado Boulder

Connor Arnold | Tristan Brasov | Alexa DeMuth | Ashwin Francis | Jacob Jones
Robert Kaplan | Zain Parkar | Abhistha Singh | Eesha Tolety | Jarod Utz | Adhindra VS

Project Overview

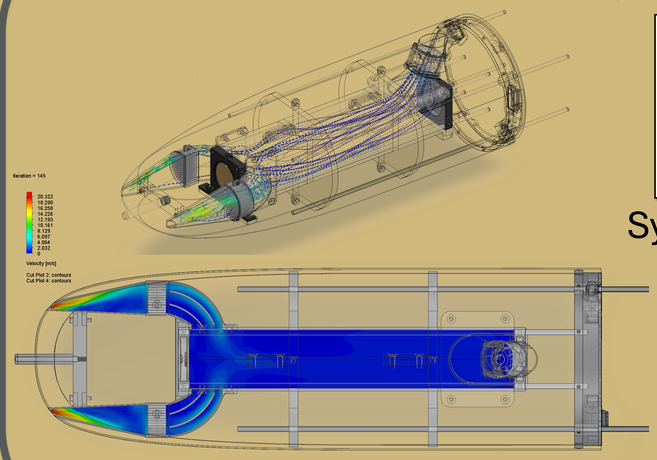


Can a fixed optical-cavity isotope instrument on a UAV accurately characterize $\delta^{18}O$ in flight?

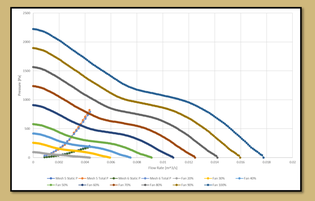


Black Swift S3 UAV

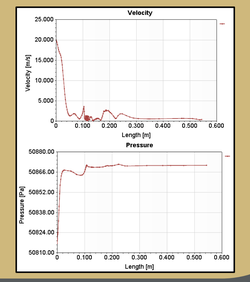
Airflow Analysis



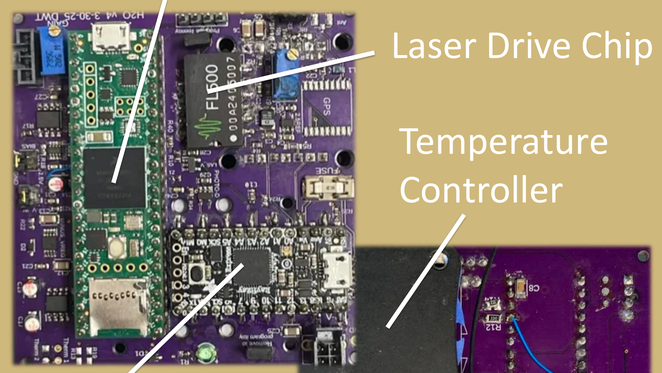
Velocity Contours

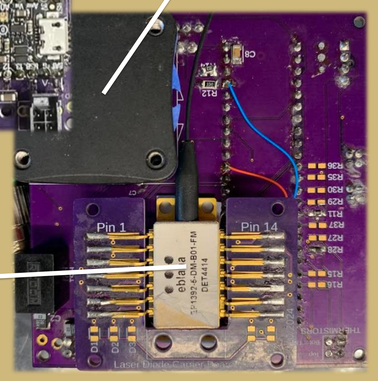


System & Fan Curve

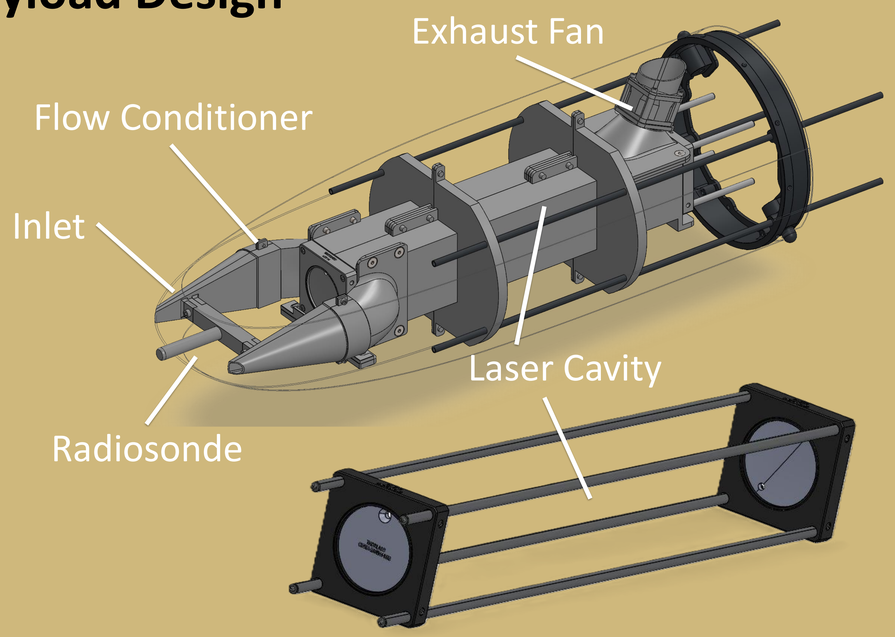


Payload Design



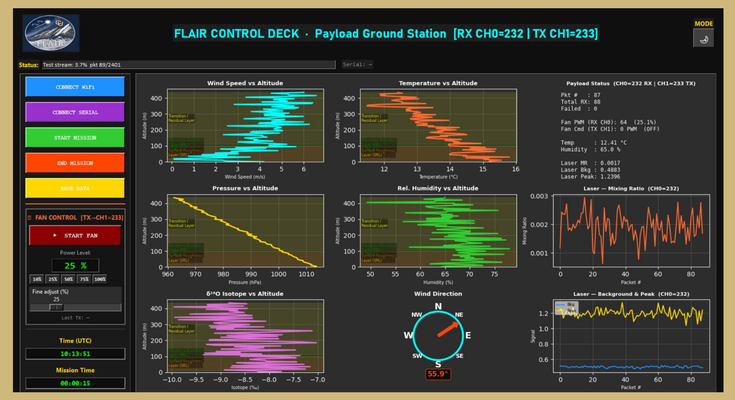


PCB Layout



CAD Model & Laser Cavity Inside Nose Cone

Software



In Flight Data Recording Software

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CHECK OUT OUR RESUMES!